

Prajakth N Kumar

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Profile

Information Science undergraduate with strong foundations in data structures, algorithms, operating systems, computer networks, and database systems. Experienced in building secure backend and full-stack software systems, with research experience in machine learning. Seeking Software Development Engineer roles in backend and systems engineering.

Education

B.M.S College of Engineering VTU <i>Bachelor of Engineering in Information Science</i>	2023 – 2027 CGPA: 9.13/10
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Technical Skills

Languages: C++, Python, C, SQL
Backend Development: FastAPI, Node.js, Express.js, REST APIs
Databases: PostgreSQL, MongoDB
Tools: Git, GitHub Actions, Linux, Docker
Cloud: Azure, Google Cloud Platform (GCP)
Libraries: NumPy, SciPy, PyTorch, Matplotlib
Core CS: Data Structures & Algorithms, Object-Oriented Programming, OS, DBMS, Computer Networks

Projects

Zero Trust Banking Engine <i>FastAPI, PostgreSQL, Keycloak, OPA, React</i> GitHub	2025
- Engineered a role-based banking platform using Keycloak and Open Policy Agent (OPA) to enforce least-privilege access and policy-driven authorization. - Developed asynchronous FastAPI services with PostgreSQL, JWT authentication, and secure session management for scalable backend operations.	
Battery SOH Prediction using Physics-Informed LSTMs <i>PyTorch, NumPy, SciPy</i> GitHub	2025
- Designed and implemented a hybrid Physics-Informed LSTM framework for battery degradation modeling using the NASA battery cycling dataset. - Integrated physics-based constraints into the training pipeline, improving prediction robustness and achieving state-of-the-art battery health estimation performance.	
Skill Exchange Platform <i>React, Node.js, Express.js, MongoDB, Azure</i> GitHub	2025
- Built and deployed a full-stack platform enabling peer-to-peer skill sharing with JWT-authenticated REST APIs and MongoDB-backed persistence. - Automated deployment workflows using GitHub Actions and deployed services on Azure App Service.	

Publications

A Temperature- and Impedance-Aware LSTM-PINN Framework for Physically Consistent Battery SOH Prediction
Scientific Reports (Nature Portfolio), 2026
DOI: 10.1038/s41598-026-37850-y

Certifications

Google Cloud Career Launchpad (Cloud Engineer Track) – Google Cloud	Apr 2026
Oracle Cloud Infrastructure 2025 Certified Foundations Associate – Oracle	Oct 2025
Altair Machine Learning Professional Certification – Altair (RapidMiner)	Dec 2025

Extra-Curriculars

BMS College of Engineering <i>Junior Coordinator – Utsav & PhaseShift</i>	Bangalore, India 2024 - 2025
Coordinated logistics and digital outreach for large-scale college festivals (200+ team members) across multiple parallel event tracks.	